

Customer Segmentation Using Unsupervised ML Algorithms KMeans And DBSCAN

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Abstract— Customer segmentation plays a pivotal role in marketing practices. For a manager, understanding the target customer is essential for developing a successful marketing strategy; neglecting this aspect may result in inefficient resource allocation towards inappropriate targets. The goal of customer segmentation is to establish connections with the most profitable customers by crafting the most suitable marketing strategies. While numerous statistical techniques have been utilized for market segmentation, the impact of large datasets can significantly diminish their effectiveness. The purpose of clustering is to optimize similarity within clusters and maximize dissimilarity between them. In this Project, I employ K-means clustering as the primary method for segmentation, along with additional models to support our research outcomes. As a result, I have effectively divided customers into five clusters based on the relationship between their annual income and spending scores, concluding that those with high income and high spending scores are particularly well-suited for targeted marketing strategies.

Keywords— Clustering, KMeans, KMeans++, DBscan,

I. INTRODUCTION

In contemporary society, the escalation of consumer spending is a logical response to the rapid advancements in production capabilities. This phenomenon instills a sense of responsibility among individuals to partake in these developments through their purchases. While an increase in the variety and quantity of products can be beneficial for the market, a surge in consumer numbers may inadvertently lead to resource wastage if marketing strategies are misaligned with the target audience. Currently, many managers and marketing professionals are experimenting with diverse approaches to formulate effective market strategies. However, it is essential to recognize that customers are dynamic entities whose preferences can shift due to various influences. Although certain tactics, such as discounts, seasonal promotions, and membership programs, may yield temporary success, they can ultimately result in inefficient use of resources, both financial and operational. Therefore, it is crucial for managers to identify and understand the evolving patterns and behaviors of their customers. The retail industry, particularly shopping malls, often finds itself in a competitive race to attract a larger customer base and maximize profits. Several factors contribute to the challenges faced by malls in fulfilling their role effectively. Increased customer activity, limited shopping time, and reduced shopping frequency are significant issues. Moreover, the proliferation of similar malls within a given area leads consumers to gravitate towards those that offer superior product selection and service quality. This competitive landscape compels mall managers to

devise strategies that distinguish their offerings from those of their rivals.

In the current landscape characterized by data-driven decision-making, comprehending customer behavior is essential for organizations seeking to refine their marketing approaches and elevate customer satisfaction levels. Customer segmentation is the process of categorizing a customer base into specific groups based on common traits, which enables businesses to customize their products and communication strategies effectively. This initiative centers on the application of machine learning methodologies for customer segmentation, specifically focusing on the DBSCAN and K-Means algorithms, alongside the Elbow Method to determine the most suitable number of clusters.

The utilization of K-Means clustering facilitates the discovery of distinct groups within customer data, allowing for segmentation based on similarities in purchasing patterns and demographic information. To identify the optimal number of clusters, the Elbow Method is utilized, which visually illustrates the variance accounted for by each cluster, aiding in the decision-making process regarding segmentation.

Furthermore, the DBSCAN (Density-Based Spatial Clustering of Applications with Noise) algorithm presents an alternative strategy, particularly adept at recognizing clusters of diverse shapes and managing noise within the dataset. By integrating these techniques, this project aspires to provide a thorough analysis of customer segments, thereby enabling targeted marketing initiatives and enhancing customer interaction. Through this investigation, businesses can acquire valuable insights into their clientele, ultimately leading to more strategic decisions and increased profitability. This paper will outline the methodologies used, present the results, and explore the significance of effective customer segmentation in today's business environment.

Rest of the paper structured as follows:

- Describe literature review of the research papers
- Describes about proposed methodologies.
- Describes experimental analysis.
- Gives a conclusion and future scope.

II. LITERATURE REVIEW

In 2020 Li, Yue, Xiaoquan, Jianying Feng; Weisong Mu. was proposed this method of analysing the customer using that particle particle swarm optimization "Customer segmentation using K-means clustering and the adaptive particle swarm optimization algorithm." which was also very helpful for my project to take forward.

In 2024 Mozumder, Md Abu Sufian, Md Shujan Shak, Nasrin Sultana, Gourab Nicholas, Md Zahidur Rahman Farazi, Md Razaul Karim, Md Sayham Khan, and Md Shahriar Mahmud Bhuiyan proposed "Optimizing Customer Segmentation in the Banking Sector: A Comparative Analysis of Machine Learning Algorithms." Journal of Computer Science and Technology Studies. This was so helpful to take forward these for the customer segmentation analysis in shopping malls also.

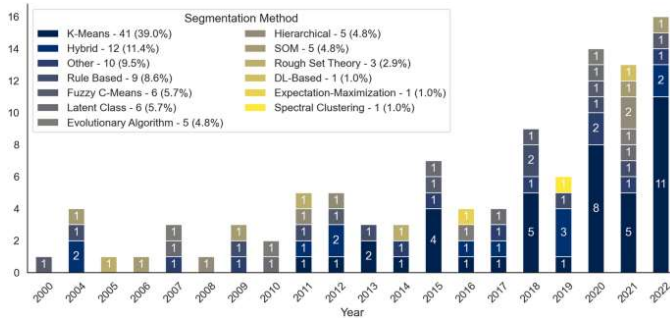


Fig 1. Dispersion of the approaches to clustering surveyed over the years of publication

According to our assessment of the literature, K-means is the most widely employed technique for consumer segmentation (41 of 105). Partitioning a collection of data points into k segments that minimize the distance between the data is the aim of the k-means algorithm. The Euclidean distance is typically applied. Numerous approximation strategies are employed since the underlying optimization problem is NP-hard to solve (MacQueen 1967; Lloyd 1982). Our k-means categorization includes the use of k-medoids and other k-means. Of the 105 relevant papers published in the from 2011 to 2022, 89 (84.76%) were published. 2011 saw the introduction of K-means (ignoring hybrid techniques). Since then, at least one application of k-means has been made annually. K-means was used in two papers in 2014, four in 2015, five in 2018, eight in 2020, and eleven in 2022. This suggests an upward trend statistically. In recent years, rule-based methods have also been employed often.

Research indicates that the identification of a customer's shopping preferences is significantly influenced by the quality of service provided and the attributes of the shopping environment. According to McGoldrick and Thompson, critical factors such as pricing, crowd density, convenience, and service quality play a pivotal role in shaping customer experiences. Furthermore, categorizing shopping malls into functional types, recreational areas, social venues, and public spaces is essential for mall managers and owners, as it aids in identifying target demographics and tailoring services accordingly.

In a different approach, Lehen and Wesley propose a classification of mall sections into various neighborhoods and value propositions. Anselmsson emphasizes that the key to achieving customer satisfaction lies in creating the right ambiance, equipping the space appropriately, offering promotions, and employing effective communication strategies. This classification aims to enhance the overall service and experience for visitors, ensuring that their needs are met in a conducive environment.

While we have discussed several critical factors that malls should consider, it is equally important to recognize the necessity of categorizing customers. This classification is vital, as misidentifying the target customer group can lead to the failure of many strategies. In this study, we will utilize machine learning techniques to classify mall customers, thereby providing a clearer visualization of existing customer segments and enhancing the effectiveness of marketing and service strategies.

III. PROPOSED CLUSTERING METHODS

The Proposed Unsupervised Machine Learning Clustering Techniques I was used to segment the different types of customers into different types of clusters as well as same types of customers into the same clusters as a result we achieve the behaviour of a customer and valuable and targeting customers to the malls and trends of the sales and in the market. In the below Fig – 2 proposed methodologies was fitted into the diagram

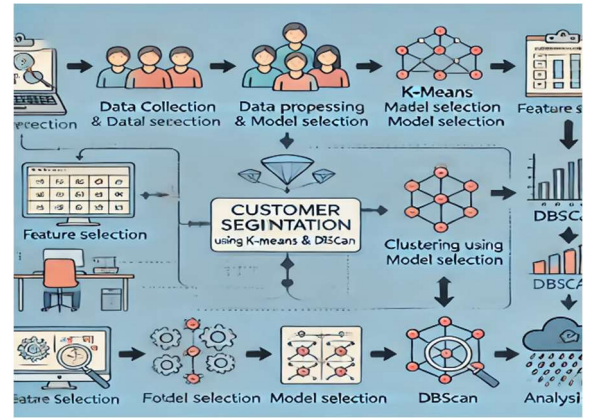


Fig 2. The workflow of the customer segmentation project

A. Data Collection

The Customer Segmentation dataset from Kaggle, created by VJ Choudhary, offers valuable insights into consumer behavior within a mall environment, making it an essential resource for understanding customer segmentation. This dataset includes several key features: CustomerID, Gender, Age, Annual Income (in thousands), and Spending Score (1-100), allowing businesses to analyze and categorize customers based on demographics and spending patterns. The significance of customer segmentation lies in its ability to enhance marketing strategies, improve customer retention, and optimize resource allocation by tailoring efforts to specific customer groups.

Analyzing this dataset typically involves data cleaning, exploratory data analysis, and the application of clustering techniques like K-Means or hierarchical clustering to identify distinct customer segments. The insights derived can inform retail strategies, product recommendations, and targeted marketing campaigns, ultimately leading to stronger customer relationships and increased sales. For those interested in exploring this dataset further, it is accessible via the Kaggle link.

B. Data Pre-processing

The data preprocessing stage is a critical component in data analysis, focusing on refining raw data to make it suitable for modeling and pattern extraction. Here, the initial steps included loading the dataset and exploring its structure. This exploration revealed columns like CustomerID, which did not contribute meaningfully to the analysis, prompting its removal. Such decisions ensure that only relevant features remain, enhancing the efficiency of subsequent analytical procedures.

Further, the dataset was scrutinized to understand the distribution of categorical variables. Specifically, the 'Gender' column was evaluated for the proportion of male and female customers, visualized through a pie chart. This visualization allowed for an intuitive grasp of gender distribution, aiding in preliminary demographic understanding. Such visual breakdowns are essential in identifying trends or imbalances that might influence subsequent clustering processes.

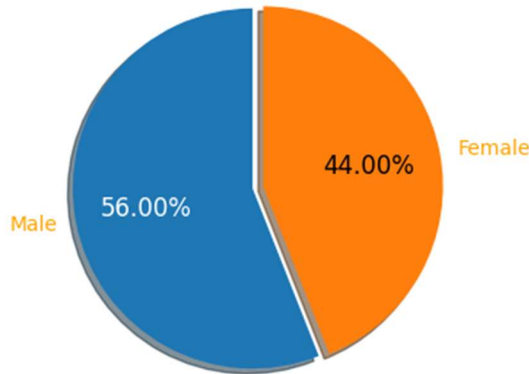


Fig 3. Phase in Data PreProcessing

The preprocessing continued with analyzing key numerical columns like 'Age,' 'Annual Income (k\$),' and 'Spending Score (1-100).' Age brackets, income groups, and spending score ranges were established to explore how customers distribute across these factors. By breaking down these variables into categories and visualizing them in bar charts, deeper insights into customer segmentation by age, income, and spending behavior emerged. These distributions provided foundational

insights that later informed clustering algorithms and ensured balanced data inputs across different value ranges.

Standardizing data was another necessary preprocessing step, especially before implementing clustering algorithms such as KMeans. By using the StandardScaler from sklearn, we normalized the dataset to have zero mean and unit variance, minimizing biases from variables with different scales. This step was crucial for ensuring that variables like 'Annual Income' and 'Spending Score,' which have vastly different units, exerted comparable influence on the clustering outcomes..

IV. MACHINE LEARNING (ML)

A. KMeans

The KMeans algorithm is one of the most commonly used clustering techniques in unsupervised learning. This algorithm aims to partition data into k clusters by minimizing the variance within each cluster. The core idea of KMeans revolves around the use of centroids to represent each cluster, which iteratively adjusts to best fit the data distribution. In our project, we applied KMeans to group customers based on attributes like 'Age' and 'Spending Score,' enabling us to identify underlying customer segments and trends..

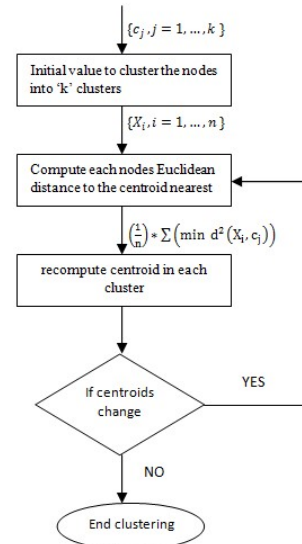
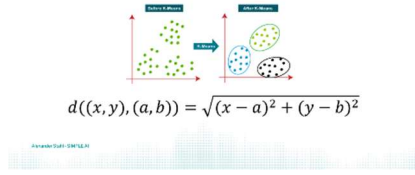


Fig 4. Detailed flowchart of KMeans Algorithm working

Euclidean Distance in KMeans

In each iteration, KMeans assigns each data point to the nearest centroid, where "nearest" is calculated using the Euclidean distance formula. This metric captures the straight-line distance between a point and a centroid in multi-dimensional space and is represented as:

Euclidean distance formula



Steps in KMeans Clustering :

The KMeans process begins with the selection of an initial set of k centroids, which serve as the core of each cluster. Each data point is assigned to the closest centroid, forming temporary clusters. The algorithm then recalculates each centroid by averaging all points assigned to it. This iterative process continues until the centroids stabilize, ensuring that each data point belongs to the most appropriate cluster. This optimization process relies on a key formula known as the **Within-Cluster Sum of Squares (WCSS)**, which minimizes the total intra-cluster variance. WCSS is expressed mathematically as:

$$WCSS = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2$$

where:

- k is the number of clusters,
- C_i is the set of points in the i -th cluster,
- x is a point in C_i ,
- μ_i is the centroid of C_i .

Choosing Optimal k : The Elbow Method

Selecting an optimal k value, or the number of clusters, is vital in ensuring meaningful clustering results. In our project, we employed the Elbow Method to determine the ideal number of clusters. This method entails plotting the WCSS values for a range of potential k values and identifying the point at which the decrease in WCSS becomes less significant. This "elbow" point signifies the optimal k , where adding more clusters no longer offers substantial improvement. The elbow method is depicted in the following steps:

1. Run KMeans for a range of k values, typically from 1 to 10.
2. Record the WCSS for each k .
3. Plot the WCSS values against the k values.
4. Identify the "elbow" point where the WCSS reduction plateaus.

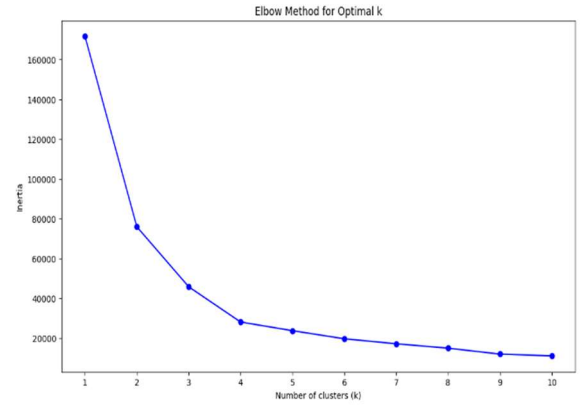


Fig 5. Visualizing Elbow chart in the project

In our dataset, we visualized the Elbow plot and found the optimal k for clustering. This approach effectively balanced clustering accuracy and simplicity.

DBSCAN: Density-Based Spatial Clustering of Applications with Noise

In contrast to KMeans, DBSCAN (Density-Based Spatial Clustering of Applications with Noise) does not require specifying the number of clusters beforehand. Instead, it clusters data based on density, allowing it to form clusters of arbitrary shape and handle noise effectively. In DBSCAN, two parameters, ϵ (the radius for neighborhood points) and min_samples (the minimum number of points required to form a dense region), determine the clustering structure. DBSCAN categorizes points into:

- Core points: Points with at least min_samples points within an ϵ radius.
- Border points: Points within ϵ of a core point but with fewer than min_samples neighbors.
- Noise points: Points that do not satisfy either condition and remain unclustered

The DBSCAN algorithm uses the distance metric to assess point proximity. In our project, applying DBSCAN yielded similar clustering outcomes to KMeans, as the core segments of customer data were dense and well-separated. The Euclidean distance formula used by DBSCAN is expressed as:

$$d(p, q) = \sqrt{\sum_{i=1}^n (p_i - q_i)^2}$$

where p and q represent two points in space.

Both KMeans and DBSCAN provided meaningful clusters in our analysis, with each technique offering unique strengths—KMeans for defined cluster counts and DBSCAN for identifying clusters of various densities. The combined results from both algorithms enriched our understanding of customer segmentation and facilitated data-driven decisions on targeting distinct customer profiles.

V. EXPERIMENTAL RESULTS

The clustering analysis of the Mall Customers dataset resulted in the formation of five distinct clusters, as shown in Figure 1. The data was grouped based on customers' age, annual income, and spending score to identify distinct consumer segments. Each color in the scatter plot represents a unique cluster, with the red dots indicating the cluster centroids. These centroids represent the "center" of each cluster, showing the typical profile for customers within each group. The clusters indicate variations in customer spending behavior and income levels across different age groups. To obtain the performance of the ML-ACO algorithm we use two evaluation metrics which are Accuracy and RMSE score. Below is the equation of these two metrics-

Silhouette Score :

$$s = \frac{b - a}{\max(a, b)}$$

The clustering model achieved a **Silhouette Score of 0.43**. This moderate score indicates that the data points are reasonably well-clustered with clear boundaries between clusters. The score suggests that most points are closer to their assigned cluster's centroid than to the centroids of other clusters, showing a satisfactory degree of separation. However, there is room for improvement, as a higher score would indicate better-defined clusters.

Davies-Bouldin Index :

$$DB = \frac{1}{k} \sum_{i=1}^k \max_{i \neq j} R_{ij}$$

The **Davies-Bouldin Index of 0.87** further validates the clustering quality, as it reflects the compactness of the clusters and the degree of separation between them. A lower Davies-Bouldin Index indicates well-separated and compact clusters, with this score suggesting that the clusters are quite distinct. This outcome is promising, given that lower values generally imply that the clusters are both cohesive and separated.

Visual Interpretation :

In the scatter plot, it is evident that the clusters exhibit a clear distribution pattern based on both age and annual income.

Certain clusters are concentrated among younger customers with lower annual incomes and spending scores, while others show higher annual incomes and spending scores among different age groups. This segmentation provides insights into potential customer targeting strategies, as each cluster may represent a different consumer behavior profile.

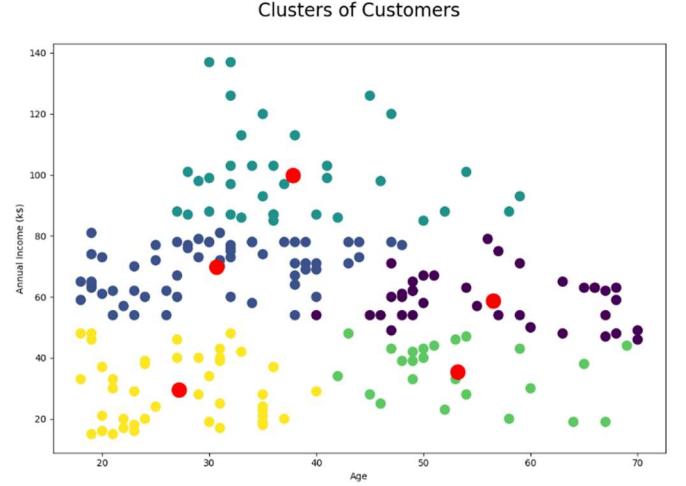


Fig 6. Visualizing Mall customers clusters

Yellow Cluster: Budget-Conscious Younger Shoppers :

The Yellow Cluster primarily includes young customers, aged 20-35, with annual incomes ranging between \$0-\$40,000. These individuals are likely early in their careers or possibly students, with limited disposable income. Consequently, they are more inclined toward budget-friendly products and often seek discounts and promotions. This cluster represents a segment that values affordability and tends to spend cautiously. To attract this group, retailers should focus on offering low-cost, trendy options along with loyalty programs and discounts.

Blue Cluster: Selective Spenders with Moderate Income :

The Blue Cluster consists of customers aged 20-40 with moderate annual incomes, generally between \$40,000-\$70,000. This group may include young professionals and small families who are selective in their purchases, focusing on both quality and value. Customers in this segment are likely to have balanced spending habits, willing to spend on items they perceive as valuable or worthwhile. Retailers targeting this group should emphasize mid-range products and consider bundle deals or occasional discounts to appeal to their value-seeking mindset.

Teal Cluster: High-Income Middle-Aged Professionals :

The Teal Cluster is characterized by customers in the age range of 30-50, with annual incomes above \$70,000. These individuals have significant spending power and may prioritize premium brands, high-quality items, and personalized services. This cluster likely includes established professionals who value exclusivity and are willing to pay for luxury. Retailers focusing on this cluster should market premium offerings and emphasize

quality and exclusivity, possibly through loyalty programs or specialized services that cater to affluent preferences.

Green Cluster: Stable, Mature Shoppers :

The Green Cluster includes customers aged 40-70 with moderate to high incomes, generally ranging from \$40,000-\$90,000. Representing a financially stable segment, these customers are mature and value quality without excessive spending. This cluster may consist of professionals approaching retirement, families with established incomes, or older individuals looking for practical and quality purchases. Retailers targeting this group should focus on products that balance quality and practicality, with an emphasis on items suited for mature lifestyles, such as lifestyle products or quality apparel.

Purple Cluster: Senior, Budget-Minded Consumers :

The Purple Cluster is composed of older customers, typically aged 50 and above, with annual incomes below \$50,000. Often on fixed incomes or retirement budgets, this segment prioritizes essential and affordable products, making them conservative spenders. They are likely looking for practicality and cost-effective options. To attract this group, retailers should emphasize affordability and highlight discounts or essential products. A focus on providing senior-specific discounts or value deals would resonate well with these budget-conscious consumers.

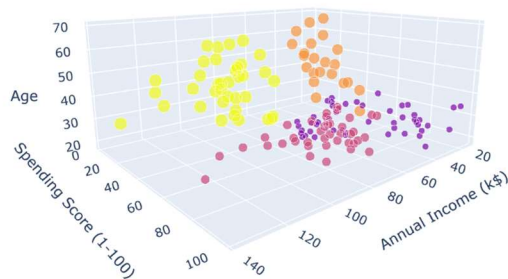


Fig 7. 3D Visualization of the clusters

The segmentation of customers into these clusters provides insights into the mall's diverse customer base, highlighting different profiles based on age, income, and spending habits. The distinct characteristics of each cluster allow for targeted marketing strategies to enhance engagement and meet the specific needs of each segment.

VI. CONCLUSION

In this study, we applied clustering algorithms to segment customers based on demographic and spending behavior data, with the goal of identifying distinct customer groups to inform targeted marketing strategies for a mall. Initial data exploration highlighted patterns in customer demographics (age and

gender) and behavior (spending score and annual income), giving insight into customer distribution across different income and spending levels. Using the KMeans algorithm, we identified distinct clusters based on the "Age" and "Spending Score" attributes. By employing the Elbow Method, we determined the optimal number of clusters, resulting in four well-defined customer groups. These clusters revealed unique characteristics such as younger, high-spending customers and middle-aged, moderate-spending groups, allowing for potential age-specific marketing campaigns. We extended this clustering to three dimensions, incorporating "Annual Income," which offered a more detailed segmentation and a foundation for income-based targeting.

The study also explored DBSCAN, a density-based clustering method, to identify natural groupings and outliers, representing customers with unusual spending or income patterns. Comparing KMeans and DBSCAN highlighted each algorithm's unique strengths, with DBSCAN effectively identifying outliers and KMeans providing clear customer clusters. Each segment identified reflects distinct customer profiles, guiding the mall's marketing strategy toward targeted promotions and tailored product recommendations. This segmentation strategy can help improve customer satisfaction, increase sales, and strengthen loyalty. Future research could incorporate additional variables like shopping frequency and product preferences to further refine the segmentation model, and explore hybrid techniques for even deeper insights. This analysis demonstrates the power of clustering in segmenting mall customers, offering valuable insights for personalized marketing.

REFERENCES

- [1] Cooil, Bruce, Lerzan Aksoy, and Timothy L. Keiningham. "Approaches to customer segmentation." *Journal of Relationship Marketing* 6, no. 3-4 (2008): 9-39.
- [2] Sari, Juni Nurma, Lukito Edi Nugroho, Ridi Ferdiana, and P. Insap Santosa. "Review on customer segmentation technique on ecommerce." *Advanced Science Letters* 22, no. 10 (2016): 3018-3022.
- [3] Kim, Su-Yeon, Tae-Soo Jung, Eui-Ho Suh, and Hyun-Seok Hwang. "Customer segmentation and strategy development based on customer lifetime value: A case study." *Expert systems with applications* 31, no. 1 (2006): 101-107.
- [4] Alves Gomes, Miguel, and Tobias Meisen. "A review on customer segmentation methods for personalized customer targeting in e-commerce use cases." *Information Systems and e-Business Management* 21, no. 3 (2023): 527-570.
- [5] Peker, Serhat, and Özge Kart. "Transactional data-based customer segmentation applying CRISP-DM methodology: A systematic review." *Journal of Data, Information and Management* 5, no. 1 (2023): 1-21.
- [6] Tabianan, Kayalvily, Shubashini Velu, and Vinayakumar Ravi. "K-means clustering approach for intelligent customer segmentation using customer purchase behavior data." *Sustainability* 14, no. 12 (2022): 7243..
- [7] KATRAGADDA, VAMSI. "Dynamic Customer Segmentation: Using Machine Learning to Identify and Address Diverse Customer Needs in Real-Time." *IRE Journals* 5, no. 10 (2022): 278-279.
- [8] Mozumder, Md Abu Sufian, Fuad Mahmud, Md Shujan Shak, Nasrin Sultana, Gourab Nicholas Rodrigues, Md Al Rafi, Md Zahidur Rahman Farazi, Md Razaul Karim, Md Sayham Khan, and Md Shahriar Mahmud Bhuiyan. "Optimizing Customer Segmentation in the Banking Sector: A

- Comparative Analysis of Machine Learning Algorithms." *Journal of Computer Science and Technology Studies* 6, no. 4 (2024): 01-07.
- [9] Saxena, Arpit, Ashi Agarwal, Binay Kumar Pandey, and Digvijay Pandey. "Examination of the Criticality of Customer Segmentation Using Unsupervised Learning Methods." *Circular Economy and Sustainability* (2024): 1-14
 - [10] Yıldız, Emre, Ceyda Güngör Şen, and Eyüp Ensar Işık. "A hyper-personalized product recommendation system focused on customer segmentation: An application in the fashion retail industry." *Journal of Theoretical and Applied Electronic Commerce Research* 18, no. 1 (2023): 571-596.
 - [11] Rungruang, Chongkolnee, Pakwan Riyapan, Arthit Intarasit, Khanchit Chuarkham, and Jirapond Muangprathub. "RFM model customer segmentation based on hierarchical approach using FCA." *Expert Systems with Applications* 237 (2024): 121449.
 - [12] Mentzer, John T., Daniel J. Flint, and G. Tomas M. Hult. "Logistics service quality as a segment-customized process." *Journal of marketing* 65, no. 4 (2001): 82-104.
 - [13] Yankelovich, Daniel, and David Meer. "Rediscovering market segmentation." *Harvard business review* 84, no. 2 (2006): 122.
 - [14] Xu, Mark, and John Walton. "Gaining customer knowledge through analytical CRM." *Industrial management & data systems* 105, no. 7 (2005): 955-971.
 - [15] Ngai, Eric WT, Li Xiu, and Dorothy CK Chau. "Application of data mining techniques in customer relationship management: A literature review and classification." *Expert systems with applications* 36, no. 2 (2009): 2592-2602.
 - [16] Ryals, Lynette. "Making customer relationship management work: the measurement and profitable management of customer relationships." *Journal of marketing* 69, no. 4 (2005): 252-261.
 - [17] Netzer, Oded, James M. Lattin, and Vikram Srinivasan. "A hidden Markov model of customer relationship dynamics." *Marketing science* 27, no. 2 (2008): 185-204.
 - [18] Ryals, Lynette, and Adrian Payne. "Customer relationship management in financial services: towards information-enabled relationship marketing." *Journal of strategic marketing* 9, no. 1 (2001): 3-27.
 - [19] Bowden, Jana Lay-Hwa. "The process of customer engagement: A conceptual framework." *Journal of marketing theory and practice* 17, no. 1 (2009): 63-74.
 - [20] Yankelovich, Daniel, and David Meer. "Rediscovering market segmentation." *Harvard business review* 84, no. 2 (2006): 122.
 - [21] Wu, Jun, Li Shi, Wen-Pin Lin, Sang-Bing Tsai, Yuanyuan Li, Liping Yang, and Guangshu Xu. "[Retracted] An Empirical Study on Customer Segmentation by Purchase Behaviors Using a RFM Model and K-Means Algorithm." *Mathematical Problems in Engineering* 2020, no. 1 (2020): 8884227.
 - [22] Wu, Shuli, Wei-Chuen Yau, Thian-Song Ong, and Siew-Chin Chong. "Integrated churn prediction and customer segmentation framework for telco business." *Ieee Access* 9 (2021): 62118-62136.